

NCC 2022



ABCB NCC 2022 VOLUME TWO (AMENDMENT 2) BASIX Thermal Comfort Energy Efficiency Assessment

Accredited Star Rating
7.0

REFERENCE
40283_v1.3

SITE ADDRESS
Lot 217 (#47) Hawk Lane ORANGE 2800

DWELLING TYPE
Single Storey

COMMISSIONED BY
G.J. Gardner Homes

ASSESSMENT DATE
15/04/2026

Energy Advance Australia Pty. Ltd.
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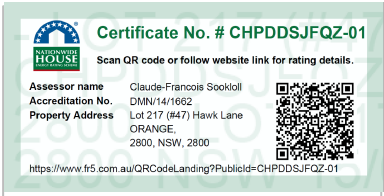
THE SUMMARY

Address	Lot 217 (#47) Hawk Lane ORANGE 2800	
Dwelling Type	Single Storey	Façade Facing
State	New South Wales	South
Site Exposure	Suburban	Garage Side (viewed from the front)
Ground Floor Type	Concrete Slab-on-Ground	Right-Hand Side
NatHERS Climate Zone	65	Modelled Wall Colour
FirstRate 5 Engine:	Chenath Engine 3.22	Wall Colour: Medium
Certificate Number	CHPDDSJFQZ-01	Modelled Roof Colour
Accredited Star Rating	7	Solar Absorptance: Medium roof colour
		Modelled Glazing Frame Colour
		Glazing Frame Colour: Medium
Conditioned Floor Area (m ²)	73.70	
Unconditioned Floor (m ²)	5.50	
Total (m ²)	79.20	

	Area (m2)	Allowance (W/m2)	Total Maximum Watts
Class 1 Total Area	90.74	5.0	453.7
Class 10a Total Area	22.54	3.0	67.6
Total Outdoor Areas	11.88	4.0	47.5
Maximum Ceiling Insulation Penetration	Maximum Allowance 0.50%	Maximum Penetration (m2) 0.45	
If approved fireproof downlight covers, which can be fully covered by insulation, are specified and noted on the electrical plan by the building designer or architect or if IC4-rated downlights are installed, then there is no need to allow for the ceiling penetration.			

ASSESSMENT CALCULATIONS & SOFTWARE RESULTS

	Target (MJ/m ² .pa)	Proposed (MJ/m ² .pa)	Efficiency Benchmark
Heating:	150.0	148.4	Pass: 1.1%
Cooling:	17.9	6.4	Pass: 94.7%
Total:	167.9	154.8	



THE ANALYSIS

DWELLING OVERVIEW

Operational Energy and Emissions Summary

This home used approximately 3,038 kWh of heating energy, equivalent to driving a petrol car about 16,693 km and producing roughly 2,400 kg of CO₂ emissions. It also used around 131 kWh of cooling energy over the year, comparable to driving 720 km and generating about 104 kg of CO₂ emissions from fuel use.

HEATING PERFORMANCE

Heating performance is 1% above the benchmark for this climate zone.
Heating efficiency meets expectations; passive heat capture and retention appear balanced.

In this cool-temperate climate zone, low heating demand confirms that insulation, glazing and winter solar design are performing effectively.

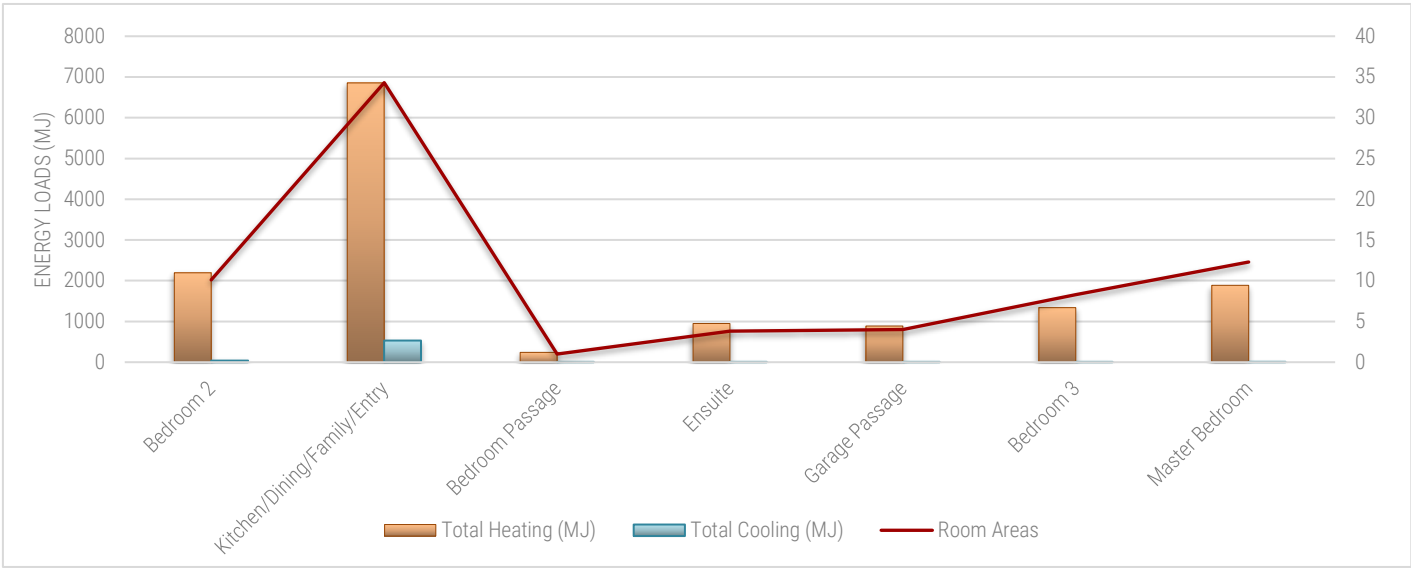
COOLING PERFORMANCE

Cooling performance is 95% above the benchmark for this climate zone.
Cooling efficiency is strong; external shading, reflective roofing and low-SHGC glazing appear to be working effectively.

In this cool-temperate climate zone, minimal cooling demand shows that shading, glazing and roof finish are well matched to local summer conditions.

Room-by-Room Energy Use Analysis

This chart shows how much energy each room needs to stay comfortable. Higher bars mean more energy needed to heat or cool that space. The line shows what's considered efficient for that room size. Bars above the line highlight where comfort is costing more energy than it should, and where upgrades can have the biggest impact.



THE ANALYSIS

GLAZING & INSULATION

Understanding the Window Ratio

The ratio below shows the percentage of the dwelling’s exterior that’s made up of glass compared to the floor area. Ideally, a lower glazing ratio supports better thermal performance, helping reduce unwanted heat loss or gain, improving comfort and lowering energy bills.

In cool temperate regions, well-insulated and well-oriented glazing improves warmth and comfort during colder months without excessive heat loss.

CAUTIONARY: Glazing size likely contributes to higher energy use; reducing window area could improve efficiency.

28.2%

This home's glazing ratio is quite high, leading to both heat loss in winter and unwanted heat gain in summer. This drives higher energy use throughout the year; reducing overall glass area would noticeably improve comfort and efficiency.

Reducing glazing on cold façades and upgrading insulation will improve winter results.

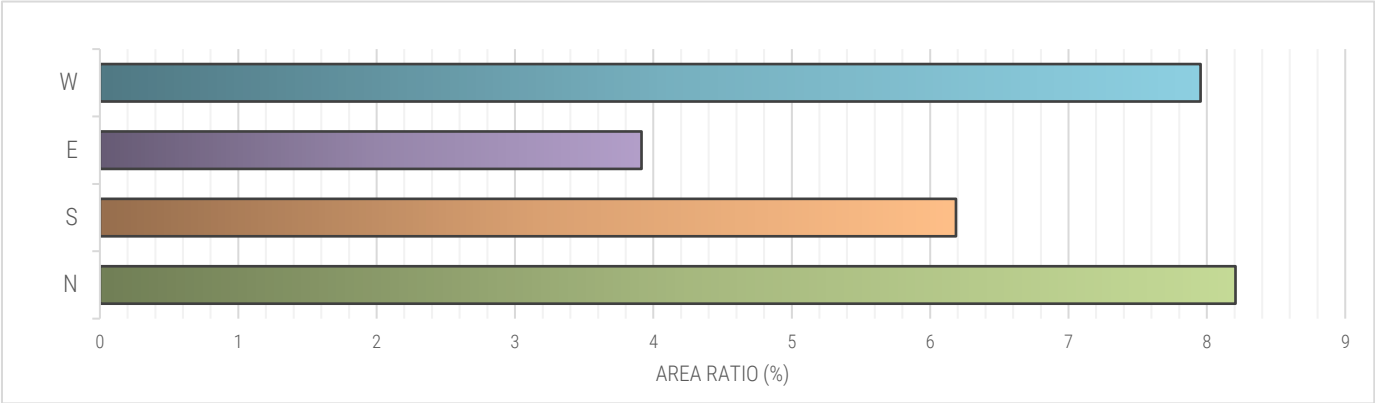
Sunlight, Heat and Glass: Orientation Overview

This dwelling’s glazing distribution varies in suitability depending on the NatHERS climate zone (cool).

Glazing to the north totals 6.5 m² (8.2% of floor area). North glazing is highly beneficial for passive winter heating; ensure adequate area and good SHGC. Glazing to the south totals 4.9 m² (6.2% of floor area). South glazing should be moderate to reduce heat loss. Glazing to the east totals 3.1 m² (3.9% of floor area). East glazing provides little winter gain and can cause summer heat; modest areas are best. Glazing to the west totals 6.3 m² (8.0% of floor area). West glazing causes summer heat and limited winter benefit; minimise or shade.

Adjusting glazing proportions—especially in directions that introduce unnecessary heat gain or loss—can significantly improve comfort and energy efficiency.

Note: NatHERS modelling includes shading and eaves. This summary explains glazing orientation impacts within the climate context.



THE ANALYSIS

ENERGY IMPACT AND EFFECTIVENESS

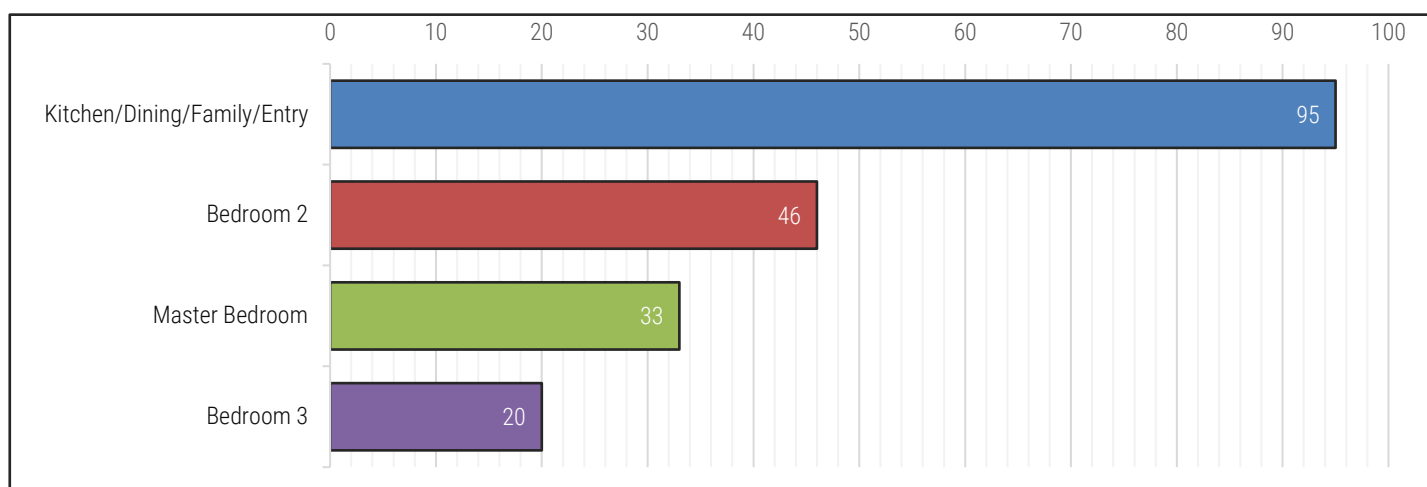
Energy Advance Comfort & Performance Impact Index

The Impact Index is Energy Advance's combined performance score for each room. It incorporates climate zone, window-to-floor ratio, heating load, cooling load, and glazing behaviour to show which rooms have the strongest influence on overall energy performance.

A higher Impact Index indicates a room where improvements will deliver greater comfort and efficiency gains.

In this home, the rooms with the highest Impact Index provide the best opportunity to improve comfort and reduce energy use through glazing or possible insulation upgrades.

Rooms with higher Impact Index scores offer the greatest "bang for buck." Upgrades in these areas have the strongest effect on overall energy performance, making them the most effective places to focus improvement efforts.



Wall Power Factor

The Wall Power Factor shows how much of your home's external shell can be improved through insulation. It compares the insulatable wall area with the total area of walls, windows and external doors.

A higher factor means insulation upgrades can meaningfully improve performance, while a lower factor indicates glazing has a greater influence than insulation.

84% Good Potential

A large share of your home's external envelope can accept upgraded insulation, offering good potential for meaningful improvements to both comfort and efficiency. Insulation upgrades help to reduce heating and cooling demand and support more consistent indoor conditions throughout the year.

While glazing does affect thermal performance, upgraded wall insulation remains an important pathway to improving overall energy efficiency and achieving a higher star rating.

Note: R2.7 insulation is already installed in parts of the home. Increasing insulation beyond this level would require increasing the wall cavity depth beyond 90 mm. Please confirm whether this construction change is feasible.

THE SPECIFICATIONS

Walls

CONSTRUCTION TYPE AND INSULATION	FRAME TYPE	LOCATION	THERMAL BREAK?
WDF Framed R2.7 Insulation VP Wrap	Timber	External	N/A
WDF Brick Veneer R2.7 Insulation VP Wrap	Timber	External	N/A
WDF INT Plasterboard Stud Wall R2.7 Insulation No Wrap	Timber	Internal	N/A
OTH Double Leaf Brick Wall 110mm Uninsulated	None	External	None

ADDITIONAL NOTES

Internal wall insulation: Throughout all internal walls
Wall Colour: Medium

Roof and Ceiling

CONSTRUCTION TYPE	CEILING INSULATION (R)	SARKING	THERMAL BREAK?	BLANKET (R)
Metal Roof OR Tiled Roof w/Sarking	7.0	No	No	1.3
Metal Roof OR Tiled Roof w/Sarking	2.5	No	No	1.3

ADDITIONAL NOTES

Solar Absorptance: Medium roof colour
Ceiling Insulation to the House and Garage area | R2.5 Perimeter Batts included

Floor

CONSTRUCTION TYPE	VENTILATION	FLOOR INSUL (R)	SLAB EDGE (R)	FLOOR AREAS (m ²)
85mm Concrete 225mm Waffle	Enclosed	Integrated	None	99.1

ADDITIONAL NOTES

Glazing

Glazing Frame Colour: Medium						
WERS CODE*	CHARACTERISTIC	TYPE	U _w -VALUE	SHGC _w	AREA (m ²)	AS-BUILT GLAZING TYPES
LID-012-06 A	Double Glazing+SGL LowE	Awning Window	3.78	0.48	15.30	Awning Windows
ALM-001-01 A	Standard Single Glazing	Awning Window	6.70	0.57	1.67	Entry Door Glazing
AWS-013-70 A	Double Glazing+SGL LowE	Hinged Door	3.09	0.49	3.78	Sliding Doors



THE REGULATIONS

13.7.1 Application of Part 13.7

- (1) This Part applies to (a) a Class 1 building, (b) a Class 10a building, and (c) a Class 10b swimming pool associated with a Class 1 or 10a building.
(2) Part 13.7 must be applied as directed in H6D2(2).

13.7.2 Insulation of services

Thermal insulation for central heating water piping and heating and cooling ductwork must—

- (a) be protected against the effects of weather and sunlight, and (b) be able to withstand the temperatures within the piping or ductwork; and (c) use thermal insulation material by AS/NZS 4859.1.

13.7.3 Central heating water piping

- (1) Central heating water piping that is not within a conditioned space must be thermally insulated to achieve the minimum material R-Values as set out in (2) to (6).
(2) Internal piping including— (a) flow and return piping that is— (i) within an unventilated wall space or (ii) within an internal floor between storeys; or (iii) between ceiling insulation and a ceiling and (b) heated water piping encased within a concrete floor slab (except that which is part of a floor heating system), must, in all climate zones, have a minimum material R-Value of 0.4.
(3) Piping located within a ventilated wall space, an enclosed building subfloor or a roof space, including— (a) flow and return piping; and (b) cold water supply piping within 500 mm of the connection to the central water heating system; and (c) relief valve piping within 500 mm of the connection to the central water heating system, must have a minimum material R-Value by (5).
(4) Piping located outside the building or in an unenclosed building subfloor or roof space, including— (a) flow and return piping; and (b) cold water supply piping within 500 mm of the connection to the central water heating system; and (c) relief valve piping within 500 mm of the connection to the central water heating system, must have a minimum material R-Value by (6).
(5) Piping referred to in (3) must have a minimum material R-Value of— (a) in climate zones 1, 2, 3 and 5 — 0.6; and (b) in climate zones 4, 6 and 7 — 0.9; and (c) in climate zone 8 — 1.3.
(6) Piping referred to in (4) must have a minimum material R-value of— (a) in climate zones 1, 2, 3 and 5 — 0.6; and (b) in climate zones 4, 6 and 7 — 1.3; and (c) in climate zone 8 — 1.3.

13.7.4 Heating and cooling ductwork

- (1) Heating and cooling ductwork and fittings must— (a) achieve the material R-Value in (4), and (b) be sealed against air loss— (i) by closing all openings in the surface, joints and seams of ductwork with adhesives, mastics, sealants or gaskets by AS 4254.1 and AS 4254.2 for a Class C seal; or (ii) for flexible ductwork, with a draw band in conjunction with a sealant or adhesive tape.
(2) Duct insulation must— (a) abut adjoining duct insulation to form a continuous barrier and (b) be installed so that it maintains its position and thickness, other than at flanges and supports; and where located outside the building, under a suspended floor, in an attached Class 10a building or in a roof space— (i) be protected by an outer sleeve of protective sheeting to prevent the insulation becoming damp, and (ii) have the outer protective sleeve sealed with adhesive tape not less than 48 mm wide creating an airtight and waterproof seal.
(3) The requirements of (1) do not apply to heating and cooling ductwork and fittings located within the insulated building envelope including a service riser within the conditioned space, internal floors between storeys and the like.
(4) The material R-Value required by (1)(a) must be determined by the following: (a) In a heating-only system or cooling-only system including an evaporative cooling system— (i) ductwork must have a minimum material R-Value of— (A) in climate zones 1 to 7 — 1.0; and (B) in climate zone 8 — 1.5; and (ii) fittings must have a minimum material R-Value of 0.4.
(b) In a combined heating and refrigerated cooling system— (i) ductwork must have a minimum material R-Value of— (A) in climate zones 1, 3, 4, 6 and 7 — 1.5; and (B) in climate zones 2 and 5 — 1.0; and (C) in climate zone 8 — 1.5; and (ii) fittings must have a minimum material R-Value of 0.4.
(c) For (b)(i), the minimum material R-value required for ductwork may be reduced by 0.5 for combined heating and refrigerated cooling systems in climate zones 1, 3, 4, 6 and 7 if the ducts are— (i) under a suspended floor with an enclosed perimeter; or in a roof space that has an insulation of greater than or equal to R0.5 directly beneath the roofing.

THE REGULATIONS

13.7.5 Electric resistance space heating

An electric resistance space heating system that serves more than one room must have— (a) separate isolating switches for each room and (b) a separate temperature controller and time switch for each group of rooms with common heating needs and (c) power loads of not more than 110 W/m² for living areas, and 150 W/m² for bathrooms.

13.7.6 Artificial lighting

(1) The lamp power density or illumination power density of artificial lighting, excluding heaters that emit light, must not exceed the allowance of— (a) 5 W/m² in a Class 1 building and (b) 4 W/m² on a Verandah, balcony or the like attached to a Class 1 building; and (c) 3 W/m² in a Class 10a building associated with a Class 1 building.

(2) The illumination power density allowance in (1) may be increased by dividing it by the relevant illumination power density adjustment factor for a control device in (6) as applicable.

(3) When designing the lamp power density or illumination power density, the power of the proposed installation must be used rather than nominal allowances for exposed batten holders or luminaires.

(4) If halogen lamps are installed, they must be separately switched from fluorescent lamps.

(5) Artificial lighting around the perimeter of a building must— (a) be controlled by a daylight sensor or (b) have an average light source efficacy of not less than 40 Lumens/W.

(6) The following illumination power density adjustment factors apply to control devices for artificial lighting:

(a) Lighting timer for corridor lighting: 0.7. Motion detector — (i) (A) at least 75% of the area of space is controlled by one or more motion detectors; or

(b) an area of less than 200 m² is switched as a block by one or more motion detectors; and (i) 0.7, where up to 6 lights are switched as a block by one or more detectors; and (ii) 0.55, where up to 2 lights are switched as a block by one or more detectors.

(c) Manual dimming system where not less than 75% of the space area is controlled by manually operated dimmers: 0.85.

(d) Programmable dimming system where not less than 75% of the space area is controlled by programmable dimmers: 0.85.

(e) Dynamic dimming system, with automatic compensation for lumen depreciation, the design lumen depreciation factor is not less than — (i) 0.9 for fluorescent lights or (ii) 0.8 for high-pressure discharge lights.

(f) Fixed dimming where at least 75% of the area is controlled by fixed dimmers that reduce the overall lighting level and the power consumption of the lighting — equal to the % of full power to which the dimmer is set divided by 0.95.

(g) Daylight sensor and dynamic lighting control device, with dimmed or stepped switching of lights adjacent to windows: (i) Lights within the space adjacent to windows other than roof lights for a distance from the window equal to the depth of the floor at window head height: 0.5. (ii) Lights within the space adjacent to roof lights: 0.6.

(7) For (6)(c), manual dimming is where lights are controlled by a knob, slider, or other mechanism or where there are pre-selected scenes that are manually selected.

(8) For (6)(d), programmed dimming is where pre-selected scenes or levels are automatically selected by the time of day, photoelectric cell, or occupancy sensor.

(9) For (6)(e), dynamic dimming is where the lighting level is varied automatically by a photoelectric cell to either proportionately compensate for the availability of daylight or the lumen depreciation of the lamps.

(10) For (6)(f), fixed dimming is where lights are controlled to a level, and that level cannot be adjusted by the user.

(11) For (6)(g)(i) and (ii), the illumination power density adjustment factor is only applied to lights controlled by that item — this adjustment factor does not apply to tungsten halogen or other incandescent sources.

13.7.7 Water heater in a heated water supply system

A water heater in a heated water supply system must be designed and installed by Part B2 of NCC Volume Three — Plumbing Code of Australia.

13.7.8 Swimming pool heating and pumping

(1) Heating for a swimming pool must be by— (a) a solar heater not boosted by electric resistance heating or

(b) a heater using reclaimed energy, (c) a gas heater, or (d) a heat pump, or (e) a combination of (a) to (d).

(2) Where some or all of the heating required by (1) is by a gas heater or a heat pump, the swimming pool must have— (a) a cover with a minimum R-Value of 0.05 unless located in a conditioned space and (b) a time switch to control the operation of the heater.

(3) A time switch must be provided to control the operation of a circulation pump for a swimming pool.

(4) For the purposes of 13.7.8, a swimming pool does not include a spa pool.

13.7.9 Spa pool heating and pumping

(1) Heating for a spa pool that shares a water recirculation system with a swimming pool must be by— (a) a solar heater or

(b) a heater using reclaimed energy or a gas heater, or (d) (e) a combination of (a) to (d).

(2) Where some or all of the heating required by (1) is by a gas heater or a heat pump, the spa pool must have— (a) a cover and (b) a push button and a time switch to control the operation of the heater.

(3) A time switch must be provided to control the operation of a circulation pump for a spa pool having a capacity of 680 L or more.

Nationwide House Energy Rating Scheme® NatHERS® Certificate No. CHPDDSJFQZ-01

Generated on 15 Apr 2026 using FirstRate5: 5.5.5a (3.22)

Property

Address Lot 217 (#47) Hawk Lane ORANGE,
2800, NSW, 2800
Lot/DP 217/1305914
NCC Class* Class 1a
Floor/all Floors
Type New Home

Plans

Main plan 40283_v1.3
Prepared by G.J. Gardner Homes

Construction and environment

Assessed floor area [m²]*	Exposure type
Conditioned* 73.7	suburban
Unconditioned* 25.5	NatHERS climate zone
Total 99.2	65 Orange AP
Garage 20	



Accredited assessor

Name Claude-Francois Sookloll
Business name Energy Advance
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Phone 1300 850 228
Accreditation No. DMN/14/1662
Assessor Accrediting Organisation
Design Matters National
Declaration of interest No

NCC Requirements

NCC provisions Volume 2
State/Territory variation Yes

National Construction Code (NCC) requirements

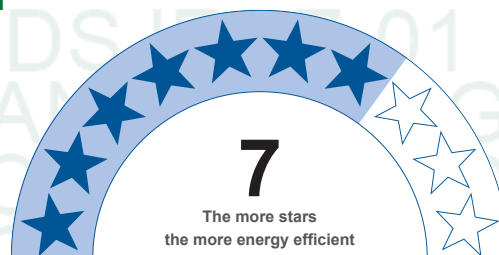
The NCC allows the use of NatHERS accredited software to comply with the energy efficiency requirements for houses (Class 1 buildings) and apartments (Class 2 sole-occupancy units and Class 4 parts of buildings). The applicable requirements for houses are detailed in Specification 42 of NCC Volume Two. For apartments the requirements are detailed in clauses J3D3 and J3D15 of NCC Volume One.

NCC 2022 includes enhanced thermal performance requirements for houses and apartments. It also includes a new whole-of-home annual energy use budget which applies to the major equipment in the home.

The NCC, and associated ABCB Standards and support material, can be accessed at www.abcb.gov.au.

Note, variations and additions to the NCC energy efficiency requirements may apply in some states and territories.

Thermal performance star rating



**NATIONWIDE
HOUSE**
ENERGY RATING SCHEME®

154.8 MJ/m²

Predicted annual energy load for heating and cooling based on standard occupancy assumptions.

For more information on your dwelling's rating see:
www.nathers.gov.au

Thermal performance [MJ/m²]

Limits taken from ABCB Standard 2022

	Heating	Cooling
Modelled	148.4	6.4
Load limits	N/A	N/A
Features determining load limits		
Floor type (lowest conditioned area)		N/A
NCC climate zone 1 or 2		N/A
Outdoor living area		N/A
Outdoor living area ceiling fan		N/A

Whole of Home performance rating

No Whole of Home
performance rating
generated for this
certificate

Verification

To verify this certificate, scan the QR code or visit <https://www.fr5.com.au/QRCodeLandIng?PublicId=CHPDDSJFQZ-01> When using either link, ensure you are visiting www.fr5.com.au.





About the ratings

Thermal performance rating

NatHERS thermal software models the expected heating and cooling energy loads using information about the design, construction, climate and common patterns of household use. The thermal performance rating (shown as a star rating on this Certificate) does not take into account appliances, apart from the airflow impacts from ceiling fans.

Whole of Home performance rating

NatHERS Whole of Home software uses the heating and cooling energy loads combined with the energy performance of the home's appliances (heating, cooling, hot water, lighting, pool/spa pump and onsite renewable energy generation and storage) and models the expected energy value* of the whole home. The Whole of Home performance rating is shown as a score out of 100 on this Certificate.

Heating & Cooling Load Limits

Additional information

In some locations under the NCC NatHERS pathway, separate heating and cooling load limits may apply. Minimum required star ratings in northern parts of Australia may also be affected by the presence or absence of an outdoor living area and/or an outdoor living area ceiling fan. Refer to the ABCB NatHERS heating and cooling load limits Standard 2022 for details or contact the relevant local building regulating authority, noting that State and Territory variations may also apply.

Setting options:

Floor type:

- CSOG – Concrete Slab on Ground
- SF – Suspended Floor (or a mixture of CSOG and SF)
- NA – Not Applicable

NCC climate Zone 1 or 2:

- Yes
- No
- NA – not applicable

Outdoor living area:

- Yes
- No
- NA – not applicable

Outdoor living area ceiling fan:

- Yes
- No
- NA – not applicable

Predicted Whole of Home annual impact by appliance

Shows the contribution each appliance has on the home's annual energy use, greenhouse gas emissions and cost without solar

Energy use:

No Whole of Home performance assessment conducted for this certificate.

Greenhouse gas emissions:

No Whole of Home performance assessment conducted for this certificate.

Cost:

No Whole of Home performance assessment conducted for this certificate.

Graph key:



Predicted onsite renewable energy impact

No Whole of Home performance assessment conducted for this certificate.

Certificate check

The checklist covers important items impacting the dwelling's ratings.
It is recommended that the accuracy of the whole certificate is checked.

Note: The boxes indicate when and who should check each item.
It is not mandatory to complete this checklist.

	Approval stage		Construction stage		
	Assessor checked	Consent authority/ surveyor checked	Builder checked	Consent authority/ surveyor checked	Occupancy/other
Genuine certificate check					
Does this Certificate match the one available at the web address or QR code verification link on the front page?		☐	☐	☐	☐
Does the NatHERS certificate number on the NatHERS-stamped plans match the number on this Certificate?		☐	☐	☐	☐
Thermal performance check					
Windows and glazed doors					
Does the window size, opening type and location shown on the NatHERS-stamped plans or as installed match what is shown in 'Window and glazed door schedule' and 'Roof window schedule' tables on this Certificate?	☐	☐	☐	☐	☐
Does the installed windows meet the substitution tolerances (AFRC* based SHGC* and U-values*) as shown in the 'Window and glazed door type and performance' and 'Roof window type and performance' tables on this Certificate?			☐	☐	☐
External walls					
Does the external wall bulk insulation (R-value) shown on the NatHERS-stamped plans or as installed match what is shown in the External wall type table on this Certificate?	☐	☐	☐	☐	☐
Does the external wall shade (colour) match what is shown in the 'External wall type' table on this Certificate?	☐	☐	☐	☐	☐
Floor					
Does the floor insulation (R-value) shown on the NatHERS-stamped plans or as installed match what is shown in the 'Floor type' table on this certificate?	☐	☐	☐	☐	☐
Ceiling penetrations*					
Does the 'quantity' and 'type' of ceiling penetrations* (e.g. downlights, exhaust fans, etc) shown on the NatHERS-stamped plans or as installed match what is shown in the 'Ceiling penetrations' table on this Certificate?	☐	☐	☐	☐	☐
Ceiling					
Does the ceiling insulation (R-value) shown on the NatHERS-stamped plans or as installed match what is shown in the 'Ceiling type' table on this Certificate?	☐	☐	☐	☐	☐
Roof					
Does the external roof shade (colour) on the NatHERS stamped plans or as installed match what is shown in the 'Roof type' table on this Certificate?	☐	☐	☐	☐	☐
Apartment entrance doors (NCC Class 2 assessments only)					
Does the 'External Door Schedule' show apartment entrance doors? Please note that an "external door" between the modelled dwelling and a shared space, such as an enclosed corridor or foyer, should not be included in the assessment (because it overstates the possible ventilation) and would invalidate the Certificate.	☐	☐		☐	☐
Exposure*					
Has the appropriate exposure type (terrain) (shown on page 1) been applied? For example, it is unlikely that a ground-floor apartment is "exposed" or a top floor high-rise apartment is "protected".	☐	☐		☐	☐
Heating and cooling load limits*					
Do the load limits settings (shown on page 1) match the values in the ABCB Standard 2022: NatHERS heating and cooling load limits for the appropriate climate zone?	☐	☐	☐	☐	☐

Certificate check

Continued

	Approval stage		Construction stage		
	Assessor checked	Consent authority/surveyor checked	Builder checked	Consent authority/surveyor checked	

Additional NCC requirements for thermal performance (not included in the NatHERS assessment)

Thermal bridging

Does the dwelling meet the NCC requirement for thermal bridging?

☐ ☐ ☐ ☐

Insulation installation method

Has the insulation been installed according to the NCC requirements?

☐ ☐ ☐

Building sealing

Does the dwelling meet the NCC requirements for Building Sealing?

☐ ☐ ☐ ☐

Whole of Home performance check (not applicable if a Whole of Home performance assessment is not conducted)

Appliances

Does the cooling appliance/s type, location and efficiency/performance shown on the NatHERS-stamped plans or as installed match the location and minimum efficiency/performance requirements shown in the Appliance schedule on this Certificate?

☐ ☐ ☐ ☐ ☐

Does the heating appliance/s type, location and efficiency/performance shown on the NatHERS-stamped plans or installed, match the location and minimum efficiency/performance requirements shown in the 'Appliance schedule' on this Certificate?

☐ ☐ ☐ ☐ ☐

Does the hot water system type and efficiency/performance shown on the NatHERS-stamped plans or as installed match the location and minimum efficiency/performance requirements shown in the 'Appliance schedule' on this Certificate?

☐ ☐ ☐ ☐ ☐

Does the pool pump efficiency/performance shown on the NatHERS-stamped plans or as installed match the minimum efficiency/performance requirements shown in the 'Appliance schedule' on this Certificate?

☐ ☐ ☐ ☐ ☐

Does the onsite renewable energy system type, orientation and system size or generation capacity shown on the NatHERS stamped plans or installed match the 'Onsite Renewable Energy schedule' on this Certificate?

☐ ☐ ☐ ☐ ☐

Additional NCC Requirements for Services (not included in the NatHERS assessment)

Does the lighting meet the artificial lighting requirements specified in the NCC?

☐ ☐ ☐ ☐

Does the hot water system meet the additional requirements specified in the NCC?

☐ ☐ ☐ ☐

Provisional values* check

Have provisional values* been used in the assessment and, if so, are they noted in 'Additional notes' table below?

☐ ☐ ☐ ☐

Other NCC requirements

Note: This Certificate only covers the energy efficiency requirements in the NCC. Additional requirements that must also be satisfied include, but are not limited to: condensation, structural and fire safety requirements and any state or territory variations to the NCC energy efficiency requirements.

Additional notes

BCA Climate Zone 7

Please note, a non-reflective vapour permeable wall wrap has been modelled throughout the external walls of this dwelling

Eaves indicated by the 'Horizontal shading feature*' maximum projection (mm) may not be directly opposing the respective wall (i.e. some eaves may be horizontally offset)

Where applicable, an additional 150mm has been added to the projection of all 'Horizontal shading features & eaves' to account for the Gutter & Fascia Board

Where applicable, proxy WERS codes may be used to meet U-Value and SHGC limitations while maintaining compliance.

*Refer to glossary.



Proxy codes refer to performance values from a comparable product when exact test data for the selected window or glazed door is unavailable

The selected window or glazed door must have a U-Value lower than the value indicated, and the SHGC must be within +/-5% of the specified value to comply



Room schedule

Room	Zone Type	Area [m²]
Master Bedroom	bedroom	12.3
Bedroom 2	bedroom	10.1
Bedroom 3	bedroom	8.3
Ensuite	nightTime	3.8
Bath	unconditioned	5.5
Kitchen/Dining/Family/Entry	kitchen	34.3
Bedroom Passage	dayTime	1
Garage Passage	dayTime	4
Garage	garage	20

Window and glazed door type and performance

Default* windows

Window ID	Window description	Maximum U-value*	SHGC*	Substitution tolerance ranges	
				SHGC lower limit	SHGC upper limit
ALM-001-01 A	Aluminium A SG Clear	6.7	0.57	0.54	0.6

Custom* windows

Window ID	Window description	Maximum U-value*	SHGC*	Substitution tolerance ranges	
				SHGC lower limit	SHGC upper limit
LID-012-06 A	Thermaview Awning Window (413 System) DG 6.38LamClr/10/6.38CPClr	3.78	0.48	0.46	0.5
AWS-013-70 A	541/542 Al Sliding Door DG 6-10-6ClimaTech	3.09	0.49	0.47	0.51

Window and glazed door schedule

Location	Window ID	Window no.	Height [mm]	Width [mm]	Window type	Opening %	Orientation	Window shading device*
Master Bedroom	LID-012-06 A	Opening 17	1500	1800	awning	60.0	N	No
Bedroom 2	LID-012-06 A	Opening 12	1800	900	awning	60.0	S	No
Bedroom 2	LID-012-06 A	Opening 13	1800	900	awning	60.0	S	No
Bedroom 3	LID-012-06 A	Opening 20	1200	1200	awning	45.0	E	No
Ensuite	LID-012-06 A	Opening 18	900	900	awning	90.0	E	No
Bath	LID-012-06 A	Opening 19	900	900	awning	90.0	E	No
Kitchen/Dining/- Family/Entry	LID-012-06 A	Opening 14	1500	1800	awning	60.0	W	No



Kitchen/Dining/- Family/Entry	LID-012-06 A	Opening 15	1500	1800	awning	60.0	W	No
Kitchen/Dining/- Family/Entry	LID-012-06 A	Opening 16	1500	600	awning	900.0	W	No
Kitchen/Dining/- Family/Entry	ALM-001-01 A	Opening 22	2040	820	casement	90.0	S	No
Kitchen/Dining/- Family/Entry	AWS-013-70 A	Opening 21	2100	1800	sliding	45.0	N	No

Roof window* *type and performance value*

Default* roof windows

Window ID	Window description	Maximum U-value*	SHGC*	Substitution tolerance ranges	
				SHGC lower limit	SHGC upper limit
No Data Available					

Custom* roof windows

Window ID	Window description	Maximum U-value*	SHGC*	Substitution tolerance ranges	
				SHGC lower limit	SHGC upper limit
No Data Available					

Roof window* *schedule*

Location	Window ID	Window no.	Opening %	Area [m²]	Width [mm]	Orientation	Outdoor shade	Indoor shade
No Data Available								

Skylight* *type and performance*

Skylight ID	Skylight description	Skylight shaft reflectance
No Data Available		

Skylight* *schedule*

Location	Skylight ID	Skylight No.	Skylight shaft length [mm]	Area [m²]	Orient- ation	Outdoor shade	Diffuser
No Data Available							

External door *schedule*

Location	Height [mm]	Width [mm]	Opening %	Orientation
Garage	2040	820	100.0	N
Garage	2250	2400	100.0	S

External wall *type*

Wall ID	Wall type	Solar absorptance	Wall shade [colour]	Bulk insulation [R-value]	Reflective wall wrap*
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1	NCC 2022 VAPOUR - TIMBER - WDF Framed R2.7 Insulation VP Wrap	0.5	Medium	Glass fibre batt (k = 0.044 density = 12 kg/m3) (R2.7)	No
2	NCC 2022 VAPOUR - TIMBER - WDF Brick Veneer R2.7 Insulation VP Wrap	0.5	Medium	Glass fibre batt (k = 0.044 density = 12 kg/m3) (R2.7)	No
3	NCC 2022 MISC - OTH Double Leaf Brick Wall 110mm Uninsulated	0.5	Medium		No

External wall *schedule*

Location	Wall ID	Height [mm]	Width [mm]	Orientation	Horizontal shading feature* maximum projection [mm]	Vertical shading feature* (yes/no)
Master Bedroom	1	2440	2990	E	711	Yes
Master Bedroom	2	2440	3890	N	600	Yes
Master Bedroom	2	2440	3580	W	3170	Yes
Bedroom 2	2	2440	1986	W	2000	Yes
Bedroom 2	2	2440	914	W	600	Yes
Bedroom 2	2	2440	2880	S	600	No
Bedroom 2	2	2440	1130	E	600	Yes
Bedroom 3	1	2440	3290	E	711	Yes
Ensuite	1	2440	1380	E	711	Yes
Bath	1	2440	2190	E	711	Yes
Kitchen/Dining/Family/Entry	2	2440	10770	W	600	Yes
Kitchen/Dining/Family/Entry	2	2440	1310	S	2350	Yes
Kitchen/Dining/Family/Entry	2	2440	2570	N	4270	Yes
Garage	2	2526	1040	N	600	Yes
Garage	3	2526	3130	S	600	Yes
Garage	2	2526	6000	E	600	Yes

Internal wall *type*

Wall ID	Wall type	Area [m²]	Bulk insulation
1	NCC 2022 STANDARD - TIMBER - WDF INT Plasterboard Stud Wall R2.7 Insulation No Wrap	84.5	Glass fibre batt (k = 0.044 density = 12 kg/m3) (R2.7)

Floor *type*

Location	Construction	Area [m²]	Sub-floor ventilation	Added insulation [R-value]	Covering
Master Bedroom	FLOOR - 85mm Concrete 225mm Waffle	9.3	Enclosed	R0.0	Carpet



Master Bedroom	FLOOR - 85mm Concrete 225mm Waffle	3	Enclosed	R0.0	Carpet
Bedroom 2	FLOOR - 85mm Concrete 225mm Waffle	2	Enclosed	R0.0	Carpet
Bedroom 2	FLOOR - 85mm Concrete 225mm Waffle	8.1	Enclosed	R0.0	Carpet
Bedroom 3	FLOOR - 85mm Concrete 225mm Waffle	6.8	Enclosed	R0.0	Carpet
Bedroom 3	FLOOR - 85mm Concrete 225mm Waffle	1.5	Enclosed	R0.0	Carpet
Ensuite	FLOOR - 85mm Concrete 225mm Waffle	3.1	Enclosed	R0.0	Tiles
Ensuite	FLOOR - 85mm Concrete 225mm Waffle	0.6	Enclosed	R0.0	Tiles
Bath	FLOOR - 85mm Concrete 225mm Waffle	4.5	Enclosed	R0.0	Tiles
Bath	FLOOR - 85mm Concrete 225mm Waffle	1	Enclosed	R0.0	Tiles
Kitchen/Dining/F- amily/Entry	FLOOR - 85mm Concrete 225mm Waffle	30	Enclosed	R0.0	Vinyl
Kitchen/Dining/F- amily/Entry	FLOOR - 85mm Concrete 225mm Waffle	4.3	Enclosed	R0.0	Vinyl
Bedroom Passage	FLOOR - 85mm Concrete 225mm Waffle	1	Enclosed	R0.0	Vinyl
Garage Passage	FLOOR - 85mm Concrete 225mm Waffle	4	Enclosed	R0.0	Vinyl
Garage	FLOOR - 85mm Concrete 225mm Waffle	2.1	Enclosed	R0.0	Tiles
Garage	FLOOR - 85mm Concrete 225mm Waffle	17.9	Enclosed	R0.0	none

Ceiling type

Location	Construction material/type	Bulk insulation R-value [may include edge batt values]	Reflective wrap*
Master Bedroom	Plasterboard	R7.0	Yes
Master Bedroom	Plasterboard	R2.5	Yes
Bedroom 2	Plasterboard	R2.5	Yes
Bedroom 2	Plasterboard	R7.0	Yes
Bedroom 3	Plasterboard	R7.0	Yes
Bedroom 3	Plasterboard	R2.5	Yes
Ensuite	Plasterboard	R7.0	Yes
Ensuite	Plasterboard	R2.5	Yes
Bath	Plasterboard	R7.0	Yes
Bath	Plasterboard	R2.5	Yes
Kitchen/Dining/F- amily/Entry	Plasterboard	R7.0	Yes



Kitchen/Dining/F- amily/Entry	Plasterboard	R2.5	Yes
Bedroom Passage	Plasterboard	R7.0	Yes
Garage Passage	Plasterboard	R7.0	Yes
Garage	Plasterboard	R7.0	Yes
Garage	Plasterboard	R7.0	Yes

Ceiling penetrations*

Location	Quantity	Type	Height [mm]	Width [mm]	Sealed/unsealed
Master Bedroom	2	Downlights	0	0	Sealed
Bedroom 2	1	Downlights	0	0	Sealed
Bedroom 3	1	Downlights	0	0	Sealed
Ensuite	1	Downlights	0	0	Sealed
Ensuite	1	Exhaust Fans	250	250	Sealed
Bath	1	Downlights	0	0	Sealed
Bath	1	Exhaust Fans	250	250	Sealed
Kitchen/Dining/Family/Ent- ry	6	Downlights	0	0	Sealed
Kitchen/Dining/Family/Ent- ry	1	Exhaust Fans	250	250	Sealed
Garage Passage	1	Downlights	0	0	Sealed
Garage	1	Downlights	0	0	Sealed
Garage	1	Exhaust Fans	250	250	Sealed

Ceiling fans

Location	Quantity	Diameter [mm]
Master Bedroom	1	1200
Bedroom 2	1	1200
Bedroom 3	1	1200
Kitchen/Dining/Family/Entry	1	1200

Roof type

Construction	Added insulation [R-value]	Solar absorptance	Roof shade [colour]
Cont:Attic-Continuous	1.3	0.73	Dark

Thermal bridging *schedule for steel frame elements*

Building element	Steel section dimensions [height x width, mm]	Frame spacing [mm]	Steel thickness [BMT,mm]	Thermal break [R-value]
No Data Available				

Appliance *schedule*

(not applicable if a Whole of Home performance assessment is not conducted for this certificate)



Note: A flat assumption of 5W/m2 is used for lighting, therefore lighting is not included in the appliance schedule.

Cooling system

Appliance/ system type	Location	Fuel type	Minimum efficiency/ performance	Recommended capacity
No Whole of Home performance assessment conducted for this certificate.				

Heating system

Appliance/ system type	Location	Fuel type	Minimum efficiency/ performance	Recommended capacity
No Whole of Home performance assessment conducted for this certificate.				

Hot water system

Appliance/ system type	Fuel type	Minimum efficiency/ performance	Hot Water CER Zone	Zone 3 STC	Assessed daily load
No Whole of Home performance assessment conducted for this certificate.					

Pool/spa equipment

Appliance/ system type	Fuel type	Minimum efficiency/ performance	Recommended capacity
No Whole of Home performance assessment conducted for this certificate.			

Onsite renewable energy *schedule*

(not applicable if a Whole of Home performance assessment is not conducted for this certificate)

System type	Orientation	System size or generation capacity
No Whole of Home performance assessment conducted for this certificate.		

Battery *schedule*

(not applicable if a Whole of Home performance assessment is not conducted for this certificate)

System type	Size [battery storage capacity]
No Whole of Home performance assessment conducted for this certificate.	

Explanatory Notes

About this report

NatHERS ratings are a reliable guide for comparing different dwelling designs and to demonstrate that designs meet the energy efficiency requirements in the National Construction Code.

NatHERS ratings use computer modelling to evaluate a home's energy efficiency and performance. They use localised climate data and standard assumptions on how people use their home to predict the heating and cooling energy loads and energy value* of the whole home. The thermal performance star rating uses the home's building specifications, layout, orientation and fabric (i.e. walls, windows, floors, roofs and ceilings) to predict the heating and cooling energy loads. The Whole of Home performance rating uses information about the home's appliances and onsite energy generation and storage to estimate the homes energy value*.

The actual energy loads, cost and greenhouse gas emissions of a home may vary from that predicted. This is because the assumptions will not always match the actual occupant usage patterns. For example, the number of occupants and how people use their appliances will vary. Energy efficient homes use less energy, are warmer on cool days, cooler on hot days and cost less to run.

Accredited assessors

For quality assured NatHERS Certificates, always use an accredited or licenced assessor registered with an Assessor Accrediting Organisation (AAO). AAOs have strict quality assurance processes, and professional development requirements ensuring consistently high standards for assessments.

Non-accredited assessors (Raters) have no ongoing training requirements and are not quality assured.

Any queries about this report should be directed to the assessor. If the assessor is unable to address questions or concerns, contact the AAO specified on the front of this certificate.

Disclaimer

The NatHERS Certificate format is developed by the NatHERS Administrator. However, the content in the certificate is entered by the assessor. It is the assessor's responsibility to use NatHERS accredited software correctly and follow the NatHERS Technical Note to produce a NatHERS Certificate.

The predicted annual energy load, cost and greenhouse gas emissions in this NatHERS Certificate are an estimate based on an assessment of the dwelling's design by the assessor. It is not a prediction of actual energy use, cost or emissions. The information and ratings may be used to compare how other dwellings are likely to perform when used in a similar way.

Information presented in this report relies on a range of standard assumptions (both embedded in NatHERS accredited software and made by the assessor who prepared this report), including assumptions about occupancy, behaviour, appliance performance, indoor air temperature and local climate.

Not all assumptions made by the assessor using the NatHERS accredited software tool are presented in this report and further details or data files may be obtained from the assessor.

Glossary

Annual energy load	the predicted amount of energy required for heating and cooling, based on standard occupancy assumptions.
AFRC	Australian Fenestration Rating Council
Assessed floor area	the floor area modelled in the software for the purpose of the NatHERS assessment. Note, this may not be consistent with the floor area in the design documents.
Ceiling penetrations	features that require a penetration to the ceiling, including downlights, vents, exhaust fans, range hoods, chimneys and flues. Excludes fixtures attached to the ceiling with small holes through the ceiling for wiring, e.g. ceiling fans; pendant lights, and heating and cooling ducts.
Conditioned	a zone within a dwelling that is expected to require heating and cooling based on standard occupancy assumptions. In some circumstances it will include garages.
COP	Coefficient of performance
Custom windows	windows listed in NatHERS software that are available on the market in Australia and have a WERS (Window Energy Rating Scheme) rating.
Default windows	windows that are representative of a specific type of window product and whose properties have been derived by statistical methods.
EER	Energy Efficiency Ratio, measure of how much cooling can be achieved by an air conditioner for a single kWh of electricity input
Energy use	This is your homes rating without solar or batteries.
Energy value	The net cost to society including, but not limited to, costs to the building user, the environment and energy networks (as defined in the ABCB Housing Provisions Standard).
Entrance door	these signify ventilation benefits in the modelling software and must not be modelled as a door when opening to a minimally ventilated corridor in a Class 2 building.
Exposure category – exposed	terrain with no obstructions e.g. flat grazing land, ocean-frontage, desert, exposed high-rise unit (usually above 10 floors).
Exposure category – open	terrain with few obstructions at a similar height e.g. grasslands with few well scattered obstructions below 10m, farmland with scattered sheds, lightly vegetated bush blocks, elevated units (e.g. above 3 floors).
Exposure category – suburban	terrain with numerous, closely spaced obstructions below 10m e.g. suburban housing, heavily vegetated bushland areas.
Exposure category – protected	terrain with numerous, closely spaced obstructions over 10 m e.g. city and industrial areas.
Horizontal shading feature	provides shading to the building in the horizontal plane, e.g. eaves, verandahs, pergolas, carports, or overhangs or balconies from upper levels.
National Construction Code (NCC) Class	the NCC groups buildings by their function and use, and assigns a classification code. NatHERS software models NCC Class 1, 2 or 4 buildings and attached Class 10a buildings. Definitions can be found at www.abcb.gov.au .
Net zero home	a home that achieves a net zero energy value*.
Opening percentage	the openability percentage or operable (moveable) area of doors or windows that is used in ventilation calculations.
Provisional value	an assumed value that does not represent an actual value. For example, if the wall colour is unspecified in the documentation, a provisional value of 'medium' must be modelled. Acceptable provisional values are outlined in the NatHERS Technical Note and can be found at www.nathers.gov.au
Recommended capacity	this is the capacity or size of equipment that is recommended by NatHERS to achieve the desired comfort conditions in the zone or zones serviced. This is a recommendation and the final selection sizing should be confirmed by a suitably qualified person.
Reflective wrap (also known as foil)	can be applied to walls, roofs and ceilings. When combined with an appropriate air gap and emissivity value, it provides insulative properties.
Roof window	for NatHERS this is typically an operable window (i.e. can be opened), will have a plaster or similar light well if there is an attic space, and generally does not have a diffuser.
Shading features	includes neighbouring buildings, fences, and wing walls, but excludes eaves.
Solar heat gain coefficient (SHGC)	the fraction of incident solar radiation admitted through a window, both directly transmitted as well as absorbed and subsequently released inward. SHGC is expressed as a number between 0 and 1. The lower a window's SHGC, the less solar heat it transmits.
Skylight (also known as roof lights)	for NatHERS this is typically a moulded unit with flexible reflective tubing (light well) and a diffuser at ceiling level.

*Refer to glossary.



STCs	Small-scale Technology Certificates, certificates created by the REC registry for renewable energy technologies that may be bought and sold as part of the Small-scale Renewable Energy Scheme operated by the Clean Energy Regulatory
Thermal breaks	are materials with an R-value greater than or equal to 0.2 that must separate the metal frame from the cladding. This includes, but is not limited to, materials such as timber battens greater than or equal to 20mm thick, continuous thermal breaks such as polystyrene insulation sheeting, plastic strips or furring channels.
U-value	the rate of heat transfer through a window. The lower the U-value, the better the insulating ability.
Unconditioned	a zone within a dwelling that is assumed to not require heating and cooling based on standard occupancy assumptions.
Vertical shading features	provides shading to the building in the vertical plane and can be parallel or perpendicular to the subject wall/window. Includes privacy screens, other walls in the building (wing walls), fences, other buildings, vegetation (protected or listed heritage trees).
Window shading device	a device fixed to windows that provides shading e.g. window awnings or screens but excludes horizontal* or vertical shading features* (eg eaves and balconies)